Assignment 3: Derivative of a Gaussian Process

CSDS 611 — Advanced Topics and Methods of Data Science I

Andrei Cozma

Let $f \sim \mathrm{GP}_{\mathbb{R}}(\mu, C)$, where μ is continuously differentiable and

$$C(x, x') = \sigma^2 \exp\left(-\frac{(x - x')^2}{2\ell^2}\right), \qquad x, x' \in \mathbb{R}, \quad \sigma^2 \geq 0, \quad \ell > 0.$$

I interpret f' as a mean-square derivative: the difference quotients converge with expected squared error tending to zero. For $h \neq 0$, define the difference quotient

$$D_h(x) = \frac{f(x + h) - f(x)}{h}.$$

To prove that f' is a Gaussian process, its values at every finite collection of inputs must be jointly Gaussian. For any N and inputs $x_1, \ldots, x_N$, the vector $(D_h(x_1), \ldots, D_h(x_N))^t$ is a linear transformation of the jointly Gaussian values of f at x_i and $x_i + h$. It is therefore jointly Gaussian.

Linearity of expectation gives

$$\mu_h(x) := \mathbb{E}[D_h(x)] = \frac{\mu(x + h) - \mu(x)}{h} \xrightarrow{h \to 0} \mu'(x).$$

To show that the difference quotients converge, compare the approximations for two step sizes, h and r. For $h, r \neq 0$, expanding the covariance gives

$$\begin{aligned} R_{h,r}(x, x') &:= \mathrm{Cov}(D_h(x), D_r(x')) \\ &= \frac{C(x + h, x' + r) - C(x + h, x') - C(x, x' + r) + C(x, x')}{hr}. \end{aligned}$$

Write

$$C_\partial(x, x') := \frac{\partial^2 C(x, x')}{\partial x \, \partial x'}.$$

Since these mixed derivatives are continuous, the fundamental theorem of calculus gives

$$R_{h,r}(x, x') = \int_0^1 \int_0^1 C_\partial(x + th, x' + sr) \, ds \, dt \longrightarrow C_\partial(x, x') \qquad (h, r \to 0).$$

The integral averages the continuous mixed derivative at points approaching (x, x'), so it converges to $C_\partial(x, x')$. For each fixed x, it follows that

$$\begin{aligned} \mathbb{E}\big[|D_h(x) - D_r(x)|^2\big] &= R_{h,h}(x, x) + R_{r,r}(x, x) - 2R_{h,r}(x, x) \\ &\quad + \big(\mu_h(x) - \mu_r(x)\big)^2 \longrightarrow 0 \qquad (h, r \to 0). \end{aligned}$$

Each covariance tends to $C_\partial(x,x)$, so the first three terms cancel in the limit. Both means tend to $\mu'(x)$, so the final term also tends to zero. Thus the difference quotients are Cauchy in mean square: their expected squared difference tends to zero as both step sizes shrink. The Cauchy criterion therefore gives a mean-square limit, which I denote by $f'(x)$. This limit satisfies

$$\mathbb{E}\big[|D_h(x) - f'(x)|^2\big] \xrightarrow{h\to 0} 0.$$

The Cauchy–Schwarz inequality lets us pass the means and covariances to the limit:

$$\mathbb{E}[f'(x)] = \mu'(x), \qquad \mathrm{Cov}(f'(x), f'(x')) = C_\partial(x,x').$$

For any finite collection of inputs, the vectors of difference quotients converge in mean square to the vector of derivative values. For a finite vector, the expected squared error is the sum of the expected squared errors of its coordinates. Since each of these tends to zero, the vector also converges in mean square. The approximating vectors are Gaussian, and their mean vectors and covariance matrices converge to those calculated above. Their limit is therefore Gaussian. Since this holds for every finite collection of inputs, f' is a Gaussian process.

Differentiating the squared exponential kernel gives

$$\begin{aligned}
\frac{\partial C(x,x')}{\partial x} &= -\frac{x-x'}{\ell^2} C(x,x'),\\
\frac{\partial C(x,x')}{\partial x'} &= \frac{x-x'}{\ell^2} C(x,x'),\\
C_\partial(x,x') &= \frac{\partial}{\partial x'}\left[-\frac{x-x'}{\ell^2} C(x,x')\right]\\
&= \left(\frac{1}{\ell^2} - \frac{(x-x')^2}{\ell^4}\right) C(x,x').
\end{aligned}$$

The kernel is symmetric, and the derivative variance is $C_\partial(x,x) = \sigma^2/\ell^2$. Positive semidefiniteness follows because C_∂ is the covariance of the derivative process.

Therefore,

$$\boxed{\begin{gathered} f' \sim \mathrm{GP}_{\mathbb{R}}(\mu', C_\partial),\\ C_\partial(x,x') = \sigma^2 \left(\frac{1}{\ell^2} - \frac{(x-x')^2}{\ell^4}\right) \exp\!\left(-\frac{(x-x')^2}{2\ell^2}\right). \end{gathered}}$$